

Individual Assignment 4

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April 29, 2023

You should complete given exercises and submit them in a single zip file named as your student ID (e.g. u210023.zip) to eClass before the deadline.

Your completed assignments will not be graded, but will be marked for completion. The maximum mark for completing all assignments in this semester is 10 points.

1 TRIZ Problems

You should review following inventive problems with the help of TRIZ and identify conflicting technical parameters. Then you should convert these conflicting parameters into one of 39 TRIZ *parameters* and find their resolution from the *contradiction matrix*. Resolutions are achieved by applying one or more inventive principles out of 40 *inventive principles* into your concept solution.

1.1 Cheese grater



Figure 1: Traditional Cheese Grater

You want to make a cheese grater that can grate cheese really quickly. A simple solution is to make it bigger. However, if you make it really big, the cheese goes all over the kitchen. These two parameters are fundamentally linked: as the size of the grater decreases, control over where the cheese goes increases, but the rate at which you can grate cheese drops. A conventional approach to this problem is to find a reasonable middle ground, a compromise: a size of grater that allows fast-enough grating and a reasonable-enough ability to direct cheese.

Write down your answer in 4 steps as shown below:

1. Identify technical contradictions in your own problem domain. Set A as the parameter improved and B the parameter affected. Write your technical contradiction as a following statement, e.g:

If we apply *this method*, parameter A improves, but parameter B gets worse.

2. Find corresponding parameters for A and B from TRIZ list. Write your technical contradiction as a following statement, e.g:

$A = 3$ and $B = 23$

3. Find corresponding inventive principles from contradiction matrix. List inventive principles by their name as shown below:

- Segmentation
- Matreshka
- Composite Material

4. Propose at least 1 concept for each of the selected inventive principles. Your concept can be a hand-sketched design or a short description of the product's important qualities that address above stated technical contradiction.

After completing your concept generation process, put all alternative concepts into a decision matrix and evaluate them against several selection criteria. While evaluating your concepts, you may consider merging them into new ones. You should submit the hand-sketched design and short description of your final concept as the main deliverable of your research.

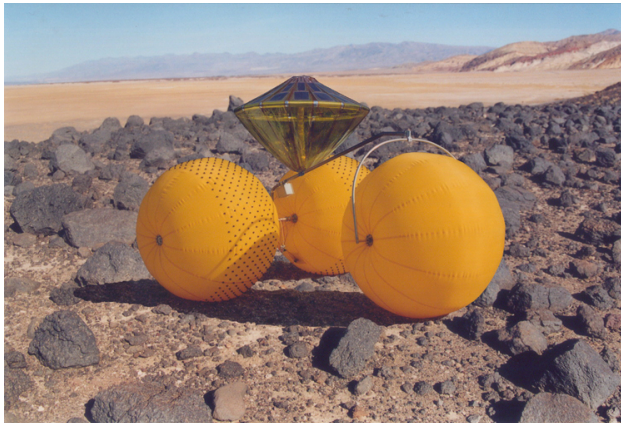


Figure 2: Mars Rover



Figure 3: Syrup filled chocolate candy

1.2 Vehicle for the planet Mars

In a science fiction story about space exploration an expedition to Mars was described. The space ship landed in a rocky valley and the astronauts promptly prepared their vehicle for a trip on the planet surface. This special vehicle was designed with big inflated tires. On the very first steep slope the vehicle tipped over. They did have small steel balls onboard to increase traction. They thought of suspending a bunch of them beneath the vehicle to lower its centre of gravity, but that would decrease the clearance and cause the weights to strike rocks and the ground more often. What can they do? Provide your answer with a brief explanation of a potential solution.

Note that they cannot change tires on Mars. Also other solutions such as partially deflating the tires to lower the centre of gravity, or installing extra tires on top and bottom sides of the vehicle, or having members of the crew lean out of the vehicle to maintain the right balance are going to cause more problems. Think about these problems too. Find an appropriate inventive principle which would help solving this problem.

1.3 Candy with a syrup inside

It was a young girl's birthday. One of the guests brought a big box of chocolate candies. The candies were shaped like small bottles filled with thick raspberry syrup. Everybody liked them. One of the guests said: "I wonder how these candies are made".

"First they made the bottles and then they filled them up with syrup," explained another guest. "The syrup would have to be very thick, otherwise the candy would not be sturdy enough," said the third guest. "At the same time, the syrup would be very difficult to pour into the bottle. It is possible to warm the syrup making it more liquid. The problem now is that the syrup would melt the chocolate bottle. We would gain in quantity and lose in quality. There would be many defective candies". How could it be done? Provide your answer with a brief explanation of a potential solution.

1.4 Steel balls inside a bent pipe

In a certain plant, one problem kept happening all the time. A bent pipe through which compressed air carried steel balls at high speed. The balls hit the inner wall of the pipe in an area of the bend. Every time a ball hit the wall, a small piece of metal chipped off. After a couple of working hours, this particular section of the pipe wore out completely, creating a hole in the thick, rugged pipe. How could you solve this problem?

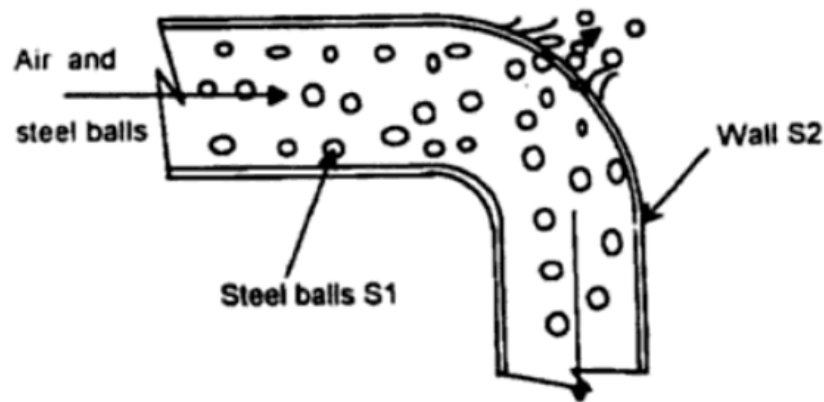


Figure 4: Steel balls inside a pipe